

LEIBSTADT, Switzerland

WMO: 066660

Lat: 47.5972N

Lon: 8.1883E

Elev: 1119

StdP: 14.11

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.5	21.8	9.9	9.5	19.6	16.4	13.1	24.5	22.8	42.4	20.4	42.5	5.5	60	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	89.5	69.3	85.8	68.4	82.1	67.0	71.4	84.4	69.8	82.2	68.2	78.9	5.9	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.4	105.1	75.1	65.9	99.6	73.0	64.6	95.2	71.8	35.8	84.6	34.4	82.8	33.1	79.0	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	14.3	12.1	DB	12.8	94.8	6.5	2.6	8.1	96.7	4.3	98.2	0.7	99.6	-4.1	101.4	(4)
(5)			WB	12.0	75.6	6.2	2.4	7.5	77.3	3.8	78.7	0.3	80.0	-4.2	81.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	34.5	35.5	43.6	51.3	57.5	64.9	68.3	67.5	60.0	51.8	42.6	35.8
	DBStd	13.58	6.51	7.28	6.11	6.29	6.35	5.93	5.61	5.74	5.49	6.29	6.16	6.06
	HDD50	1900	482	407	210	61	10	0	0	0	1	57	233	439
	HDD65	5389	946	826	664	411	244	75	31	40	166	409	673	904
	CDD50	2335	1	0	10	101	242	446	567	543	302	113	10	0
	CDD65	351	0	0	0	0	12	71	133	118	17	0	0	0
	CDH74	3593	0	0	0	27	179	785	1365	1066	169	2	0	0
	CDH80	1228	0	0	0	2	29	260	521	383	33	0	0	0
Wind	WSAvg	4.0	4.6	4.4	4.7	4.2	4.5	4.1	4.0	3.4	3.4	3.3	3.6	4.1
Precipitation	PrecAvg	52.2	4.6	4.3	3.9	3.6	4.5	4.4	4.5	4.2	3.6	4.2	4.7	5.6
	PrecMax	71.9	11.7	11.3	14.0	7.4	7.7	8.4	8.7	9.4	7.2	8.7	11.5	13.0
	PrecMin	37.3	0.6	0.9	0.8	0.2	1.4	1.7	1.2	0.7	1.4	0.2	0.2	0.0
	PrecStd	8.7	2.9	2.8	2.9	1.9	1.7	1.8	1.8	1.7	1.6	2.1	2.7	3.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	59.4	68.0	78.4	84.8	91.7	95.1	93.1	84.0	72.4	63.1	53.7
		MCWB	50.1	49.2	52.8	59.8	66.9	71.9	71.3	69.1	66.7	61.0	55.3	50.5
	2%	DB	48.8	53.1	64.1	73.1	78.8	87.0	90.0	88.5	79.1	68.0	58.1	50.4
		MCWB	45.8	45.9	50.9	56.4	62.4	69.6	68.4	69.7	65.3	59.8	53.1	48.6
	5%	DB	45.6	49.2	59.7	69.7	75.0	82.4	85.8	84.5	74.4	63.9	54.3	47.0
		MCWB	43.5	44.5	49.2	56.2	60.9	67.6	67.6	68.8	63.7	58.1	50.9	45.2
	10%	DB	42.9	45.0	55.2	65.3	70.6	78.2	81.8	80.0	70.5	60.7	50.7	44.2
		MCWB	41.1	41.5	47.5	54.1	59.5	65.5	66.5	67.2	61.9	57.0	48.6	42.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.4	50.4	54.4	60.4	68.0	74.8	72.8	73.4	68.8	63.0	56.3	52.0
		MCDB	54.2	56.1	64.0	73.8	79.3	87.4	89.3	86.0	79.5	68.9	61.3	52.2
	2%	WB	46.2	46.9	52.1	58.3	65.1	70.9	70.7	71.3	66.6	61.0	53.6	48.6
		MCDB	48.2	52.3	61.6	70.1	74.6	83.7	84.8	83.2	75.7	65.3	57.3	50.1
	5%	WB	43.8	44.5	50.0	56.3	62.5	68.7	69.1	69.7	64.7	59.3	51.2	45.6
		MCDB	45.6	48.7	57.7	67.0	71.2	79.3	81.6	81.5	72.2	62.8	53.9	46.8
	10%	WB	41.4	41.8	48.2	54.6	60.4	66.7	67.6	68.1	62.9	57.4	49.0	42.8
		MCDB	42.7	44.8	54.5	64.2	68.6	76.1	78.4	78.1	68.5	60.1	50.5	44.2
Mean Daily Temperature Range	5% DB	MDBR	8.4	12.1	17.8	20.5	19.0	20.0	21.3	20.6	18.7	14.6	9.9	8.3
		MCDBR	12.7	20.4	27.8	29.9	27.3	28.6	29.6	28.1	25.3	20.0	15.9	12.4
		MCWBR	10.0	13.7	15.9	14.9	12.9	12.8	10.5	11.2	12.5	12.2	10.9	10.1
	5% WB	MCDDBR	12.3	19.3	24.3	27.2	23.7	26.4	25.8	25.7	22.2	17.8	14.4	11.9
MCWBR		9.9	13.5	14.3	14.0	11.8	12.4	10.1	10.8	11.4	11.4	10.5	10.3	
Clear-Sky Solar Irradiance	taub	0.310	0.338	0.375	0.406	0.418	0.422	0.422	0.415	0.390	0.368	0.333	0.306	
	taud	2.429	2.341	2.281	2.226	2.237	2.272	2.265	2.299	2.347	2.385	2.435	2.432	
	Ebn at Noon	244	259	266	268	269	268	266	262	258	244	232	230	
	Edh at Noon	21	29	36	41	43	41	41	38	33	27	21	19	
All-Sky Solar Radiation	RadAvg	349	609	995	1371	1591	1840	1764	1538	1154	720	391	284	
	RadStd	41	89	141	198	161	169	160	155	117	74	47	48	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (114 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+1.31	+0.76	+0.59	N/A	N/A	+99	+25

Nomenclature: See separate page